DM545/DM871
Linear and Integer Programming

# More on Duality

Marco Chiarandini

Department of Mathematics and Computer Science

October 5, 2026

# Outline



1. Derivation
    Lagrangian Duality

2. Dual Simplex

3. Sensitivity Analysis

# Summary

- Derivation:
  1. economic interpretation
  2. bounding
  3. multipliers
  4. recipe
  5. Lagrangian
- Theory:
  - Symmetry
  - Weak duality theorem
  - Strong duality theorem
  - Complementary slackness theorem
- Economic interpretation
- Dual Simplex
- Sensitivity Analysis

# Outline

1. Derivation
    Lagrangian Duality
2. Dual Simplex
3. Sensitivity Analysis

# Outline

1. Derivation
    Lagrangian Duality
2. Dual Simplex
3. Sensitivity Analysis

# Lagrangian Duality

**Relaxation**: if a problem is hard to solve then find an easier problem resembling the original one that provides information in terms of bounds. Then, search for the strongest bounds.

$$\begin{array}{rrrrrrrrrl} \min & 13x_1 & + & 6x_2 & + & 4x_3 & +12x_4 & & & \\ & 2x_1 & + & 3x_2 & + & 4x_3 & + & 5x_4 & = & 7 \\ & 3x_1 & + & & + & 2x_3 & + & 4x_4 & = & 2 \\ & & & & & & & x_1, x_2, x_3, x_4 & \geq & 0 \end{array}$$

We wish to reduce to a problem easier to solve, ie:

$$\begin{array}{rrrrrrrrrl} \min & c_1x_1 & + & c_2x_2 & + & \ldots & +c_nx_n & & & \\ & & & & & & & x_1, x_2, \ldots, x_n & \geq & 0 \end{array}$$

solvable by inspection: if $c_j < 0$ then $x_j = +\infty$, if $c_j \geq 0$ then $x_j = 0$.

Measure of violation of the constraints:

$$\begin{array}{rrrrrrrrr} 7 & - & (2x_1 & + & 3x_2 & + & 4x_3 & + & 5x_4) \\ 2 & - & (3x_1 & + & & + & 2x_3 & + & 4x_4) \end{array}$$

We relax these measures in obj. function with **Lagrangian multipliers** $y_1$, $y_2$.
We obtain a family of problems:

$$PR(y_1, y_2) = \min_{x_1, x_2, x_3, x_4 \geq 0} \left\{ \begin{array}{rrrrrrrr} & 13x_1 & + & 6x_2 & + & 4x_3 & + & 12x_4 \\ +y_1(7- & 2x_1 & - & 3x_2 & - & 4x_3 & - & 5x_4) \\ +y_2(2- & 3x_1 & & & - & 2x_3 & - & 4x_4) \end{array} \right\}$$

1. for all $y_1, y_2 \in \mathbb{R}$ : $\mathrm{opt}(PR(y_1, y_2)) \leq \mathrm{opt}(P)$
2. $\max_{y_1, y_2 \in \mathbb{R}} \{\mathrm{opt}(PR(y_1, y_2))\} \leq \mathrm{opt}(P)$

PR is easy to solve.
(It can be also seen as a proof of the weak duality theorem)

$$PR(y_1, y_2) = \min_{x_1,x_2,x_3,x_4 \geq 0} \left\{ \begin{array}{rrrrrrr} & (13 & - & 2y_1 & - & 3y_2) & x_1 \\ + & (6 & - & 3y_1 & & ) & x_2 \\ + & (4 & - & 4y_1 & - & 2y_2) & x_3 \\ + & (12 & - & 5y_1 & - & 4y_2) & x_4 \\ + & & & 7y_1 & + & 2y_2 & \end{array} \right\}$$

if coefficient of $x$ is $< 0$ then bound is $-\infty$ then LB is useless

$$\begin{array}{rrrrrrr} (13 & - & 2y_1 & - & 3y_2) & \geq & 0 \\ (6 & - & 3y_1 & & ) & \geq & 0 \\ (4 & - & 4y_1 & - & 2y_2) & \geq & 0 \\ (12 & - & 5y_1 & - & 4y_2) & \geq & 0 \end{array}$$

If they all hold then we are left with $7y_1 + 2y_2$ because all go to 0.

$$\begin{array}{rrrrrrr} \max & 7y_1 & + & 2y_2 & & & \\ & 2y_1 & + & 3y_2 & \leq & 13 \\ & 3y_1 & & & \leq & 6 \\ & 4y_1 & + & 2y_2 & \leq & 4 \\ & 5y_1 & + & 4y_2 & \leq & 12 \end{array}$$

# General Formulation

$$\begin{array}{lrll} \min & z & = & \boldsymbol{c}^T\boldsymbol{x} & \boldsymbol{c} \in \mathbb{R}^n \\ & A\boldsymbol{x} & = & \boldsymbol{b} & A \in \mathbb{R}^{m\times n}, \boldsymbol{b} \in \mathbb{R}^m \\ & \boldsymbol{x} & \geq & \boldsymbol{0} & \boldsymbol{x} \in \mathbb{R}^n \end{array}$$

$$\max_{\boldsymbol{y}\in\mathbb{R}^m}\{\min_{\boldsymbol{x}\in\mathbb{R}^n_+}\{\boldsymbol{c}^T\boldsymbol{x} + \boldsymbol{y}^T(\boldsymbol{b} - A\boldsymbol{x})\}\}$$

$$\max_{\boldsymbol{y}\in\mathbb{R}^m}\{\min_{\boldsymbol{x}\in\mathbb{R}^n_+}\{(\boldsymbol{c}^T - \boldsymbol{y}^T A)\boldsymbol{x} + \boldsymbol{y}^T\boldsymbol{b}\}\}$$

$$\begin{array}{lrl} \max & \boldsymbol{b}^T\boldsymbol{y} & \\ & A^T\boldsymbol{y} & \leq \boldsymbol{c} \\ & \boldsymbol{y} \in \mathbb{R}^m & \end{array}$$

# Outline

1. Derivation
Lagrangian Duality

2. Dual Simplex

3. Sensitivity Analysis

# Dual Simplex



- Dual simplex (Lemke, 1954): apply the simplex method to the dual problem and observe what happens in the primal tableau:

$$\begin{aligned}\max\{\boldsymbol{c}^T\boldsymbol{x} \mid A\boldsymbol{x} \leq \boldsymbol{b}, \boldsymbol{x} \geq \boldsymbol{0}\} &= \min\{\boldsymbol{b}^T\boldsymbol{y} \mid A^T\boldsymbol{y} \geq \boldsymbol{c}^T, \boldsymbol{y} \geq \boldsymbol{0}\} \\ &= -\max\{-\boldsymbol{b}^T\boldsymbol{y} \mid -A^T\boldsymbol{y} \leq -\boldsymbol{c}^T, \boldsymbol{y} \geq \boldsymbol{0}\}\end{aligned}$$

- We obtain a new algorithm for the primal problem: the **dual simplex**
  It corresponds to the primal simplex applied to the dual

- We obtain also another way to handle infeasibility

# Primal Simplex on Dual Problem

## Example: Primal feasible start



Primal:

$$\begin{aligned} \max \quad & 4x_1 + x_2 + 3x_3 & \\ & x_1 + 4x_2 & \leq 1 \\ & 3x_1 + x_2 + x_3 & \leq 3 \\ & x_1, x_2, x_3 & \geq 0 \end{aligned}$$

- Initial tableau

| x1 | x2 | x3 | w1 | w2 | -z | b |
|---|---|---|---|---|---|---|
| 1 | 4 | 0 | 1 | 0 | 0 | 1 |
| 3 | 1 | 1 | 0 | 1 | 0 | 3 |
| 4 | 1 | 3 | 0 | 0 | 1 | 0 |

- primal feasible start

Dual:

$$\begin{aligned} \min \quad & y_1 + 3y_2 & \\ & y_1 + 3y_2 & \geq 4 \\ & 4y_1 + y_2 & \geq 1 \\ & y_2 & \geq 3 \\ & y_1, y_2 & \geq 0 \end{aligned}$$

- Initial tableau ($\min by \equiv -\max -by$)

| y1 | y2 | z1 | z2 | z3 | -p | b |
|---|---|---|---|---|---|---|
| -1 | -3 | 1 | 0 | 0 | 0 | -4 |
| -4 | -1 | 0 | 0 | 1 | 0 | -1 |
| 0 | -1 | 0 | 0 | 1 | 0 | -3 |
| -1 | -3 | 0 | 0 | 0 | 1 | 0 |

- dual infeasible start

# Primal Simplex on Dual Problem

## Example: Primal infeasible start



Primal:

$$\begin{aligned} \max \quad & -x_1 - x_2 & \\ & -2x_1 - x_2 \leq & 4 \\ & -2x_1 + 4x_2 \leq & -8 \\ & -x_1 + 3x_2 \leq & -7 \\ & x_1, x_2 \geq & 0 \end{aligned}$$

- Initial tableau

| | x1 | x2 | w1 | w2 | w3 | -z | b |
|---|---|---|---|---|---|---|---|
| | -2 | -1 | 1 | 0 | 0 | 0 | 4 |
| | -2 | 4 | 0 | 1 | 0 | 0 | -8 |
| | -1 | 3 | 0 | 0 | 1 | 0 | -7 |
| | -1 | -1 | 0 | 0 | 0 | 1 | 0 |

infeasible start

- $x_1$ enters, $w_2$ leaves

Dual:

$$\begin{aligned} \min \quad & 4y_1 - 8y_2 - 7y_3 & \\ & -2y_1 - 2y_2 - y_3 \geq & -1 \\ & -y_1 + 4y_2 + 3y_3 \geq & -1 \\ & y_1, y_2, y_3 \geq & 0 \end{aligned}$$

- Initial tableau ($\min by \equiv -\max -by$)

| | y1 | y2 | y3 | z1 | z2 | -p | b |
|---|---|---|---|---|---|---|---|
| | 2 | 2 | 1 | 1 | 0 | 0 | 1 |
| | 1 | -4 | -3 | 0 | 1 | 0 | 1 |
| | -4 | 8 | 7 | 0 | 0 | 1 | 0 |

feasible start (thanks to $-x_1 - x_2$)

- $y_2$ enters, $z_1$ leaves

- $x_1$ enters, $w_2$ leaves

| | x1 | x2 | w1 | w2 | w3 | -z | b |
|---|---|---|---|---|---|---|---|
| | 0 | -5 | 1 | -1 | 0 | 0 | 12 |
| | 1 | -2 | 0 | -0.5 | 0 | 0 | 4 |
| | 0 | 1 | 0 | -0.5 | 1 | 0 | -3 |
| | 0 | -3 | 0 | -0.5 | 0 | 1 | 4 |

- $w_2$ enters, $w_3$ leaves

| | x1 | x2 | w1 | w2 | w3 | -z | b |
|---|---|---|---|---|---|---|---|
| | 0 | -7 | 1 | 0 | -2 | 0 | 18 |
| | 1 | -3 | 0 | 0 | -1 | 0 | 7 |
| | 0 | -2 | 0 | 1 | -2 | 0 | 6 |
| | 0 | -4 | 0 | 0 | -1 | 1 | 7 |

(note that we kept $c_j < 0$, ie, optimality)

- $y_2$ enters, $z_1$ leaves

| | y1 | y2 | y3 | z1 | z2 | -p | b |
|---|---|---|---|---|---|---|---|
| | 1 | 1 | 0.5 | 0.5 | 0 | 0 | 0.5 |
| | 5 | 0 | -1 | 2 | 1 | 0 | 3 |
| | -4 | 0 | 3 | -12 | 0 | 1 | -4 |

- $y_3$ enters, $y_2$ leaves

| | y1 | y2 | y3 | z1 | z2 | -p | b |
|---|---|---|---|---|---|---|---|
| | 2 | 2 | 1 | 1 | 0 | 0 | 1 |
| | 7 | 2 | 0 | 3 | 1 | 0 | 3 |
| | -18 | -6 | 0 | -7 | 0 | 1 | -7 |

# Dual Simplex on Primal Problem

**Primal simplex on primal problem:**

1. pivot $> 0$
2. col $c_j$ with wrong sign
3. row: $\min \left\{ \frac{b_i}{a_{ij}} : a_{ij} > 0, i = 1, .., m \right\}$

**Dual simplex on primal problem:**

1. pivot $< 0$
2. row $b_i < 0$
   (condition of feasibility)
3. col: $\min \left\{ \left| \frac{c_j}{a_{ij}} \right| : a_{ij} < 0, j = 1, 2, .., n + m \right\}$
   (least worsening solution)

- Primal works with feasible solutions towards optimality
- Dual works with optimal solutions towards feasibility

# Dual Simplex



1. (primal) simplex on primal problem (the one studied so far)

2. Now: dual simplex on primal problem $\equiv$ primal simplex on dual problem
   (implemented as dual simplex, understood as primal simplex on dual problem)

Uses of the Dual Simplex:

- Infeasible start
  Dual based Phase I algorithm (**Dual-primal algorithm**)

- The dual simplex can work better than the primal in some cases.
  Eg. since running time in practice between $2m$ and $3m$, then if $m = 99$ and $n = 9$ then better the dual

# Dual based Phase I

Example:

$$\begin{aligned} \text{maximize} \quad & z = x_1 - x_2 \\ \text{subject to} \quad & x_1 + x_2 \leq 2 \\ & 2x_1 + 2x_2 \geq 2 \\ & x_1, x_2 \geq 0 \end{aligned}$$

# Summary

- Derivation:
  1. bounding
  2. multipliers
  3. recipe
  4. Lagrangian
- Theory:
  - Symmetry
  - Weak duality theorem
  - Strong duality theorem
  - Complementary slackness theorem
- Dual Simplex
- Sensitivity Analysis, Economic interpretation

# Outline

1. Derivation
Lagrangian Duality

2. Dual Simplex

3. Sensitivity Analysis

# Sensitivity Analysis
## aka Postoptimality Analysis



Instead of solving each modified problem from scratch, exploit results obtained from solving the original problem.

$$\max\{\boldsymbol{c}^T\boldsymbol{x} \mid A\boldsymbol{x} = \boldsymbol{b}, \boldsymbol{l} \leq \boldsymbol{x} \leq \boldsymbol{u}\} \qquad (*)$$

(I) changes to coefficients of objective function: $\max\{\tilde{\boldsymbol{c}}^T\boldsymbol{x} \mid A\boldsymbol{x} = \boldsymbol{b}, \boldsymbol{l} \leq \boldsymbol{x} \leq \boldsymbol{u}\}$ (primal)
$\boldsymbol{x}^*$ of (*) remains feasible hence we can restart the simplex from $\boldsymbol{x}^*$

(II) changes to RHS terms: $\max\{\boldsymbol{c}^T\boldsymbol{x} \mid A\boldsymbol{x} = \tilde{\boldsymbol{b}}, \boldsymbol{l} \leq \boldsymbol{x} \leq \boldsymbol{u}\}$ (dual)
$\boldsymbol{x}^*$ optimal feasible solution of (*)
basic sol $\bar{\boldsymbol{x}}$ of (II): $\bar{\boldsymbol{x}}_N = \boldsymbol{x}^*_N$, $A_B\bar{\boldsymbol{x}}_B = \tilde{\boldsymbol{b}} - A_N\bar{\boldsymbol{x}}_N$
$\bar{\boldsymbol{x}}$ is dual feasible and we can start the dual simplex from there. If $\tilde{\boldsymbol{b}}$ differs from $\boldsymbol{b}$ only slightly it may be we are already optimal.

(III) introduce a new variable: (primal)

$$\max \quad \sum_{j=1}^{6} c_j x_j$$

$$\sum_{j=1}^{6} a_{ij} x_j = b_i, \;\; i = 1, \ldots, 3$$

$$l_j \leq x_j \leq u_j, \;\; j = 1, \ldots, 6$$

$[x_1^*, \ldots, x_6^*]$ feasible

$$\max \quad \sum_{j=1}^{7} c_j x_j$$

$$\sum_{j=1}^{7} a_{ij} x_j = b_i, \;\; i = 1, \ldots, 3$$

$$l_j \leq x_j \leq u_j, \;\; j = 1, \ldots, 7$$

$[x_1^*, \ldots, x_6^*, 0]$ feasible

(IV) introduce a new constraint: (dual)

$$\sum_{j=1}^{6} a_{4j} x_j = b_4$$

$$\sum_{j=1}^{6} a_{5j} x_j = b_5$$

$$l_j \leq x_j \leq u_j \qquad j = 7, 8$$

$[x_1^*, \ldots, x_6^*]$ optimal

$[x_1^*, \ldots, x_6^*, x_7^*, x_8^*]$dual feasible

$$x_7^* = b_4 - \sum_{j=1}^{6} a_{4j} x_j^*$$

$$x_8^* = b_5 - \sum_{j=1}^{6} a_{5j} x_j^*$$

# Examples

(I) Variation of reduced costs:

$$\begin{array}{rrcl} \max & (6+\rho)x_1 + & 8x_2 & \\ & 5x_1 + & 10x_2 \leq & 60 \\ & 4x_1 + & 4x_2 \leq & 40 \\ & & x_1, x_2 \geq & 0 \end{array}$$

| | $x_1$ | $x_2$ | $x_3$ | $x_4$ | $-z$ | $b$ |
|---|---|---|---|---|---|---|
| $x_3$ | 5 | 10 | 1 | 0 | 0 | 60 |
| $x_4$ | 4 | 4 | 0 | 1 | 0 | 40 |
| | 6 | 8 | 0 | 0 | 1 | 0 |

The last tableau gives the possibility to estimate the effect of variations

| | $x_1$ | $x_2$ | $x_3$ | $x_4$ | $-z$ | $b$ |
|---|---|---|---|---|---|---|
| $x_2$ | 0 | 1 | $1/5$ | $-1/4$ | 0 | 2 |
| $x_1$ | 1 | 0 | $-1/5$ | $1/2$ | 0 | 8 |
| | 0 | 0 | $-2/5$ | $-1$ | 1 | $-64$ |

- For a variable in basis the perturbation goes unchanged in the red. costs. Eg:

$$\max (6+\delta)x_1 + 8x_2 \implies \bar{c}_1 = 1(6+\delta) - \frac{2}{5}\cdot 5 - 1\cdot 4 = \delta$$

  then need to bring in canonical form yielding changes in the reduced cost of other variables (might need to iterate the primal simplex) and in the obj value.

- For a variable not in basis, if it changes the sign of the reduced cost $\implies$ worth bringing in basis $\implies$ the $\delta$ term propagates to other columns

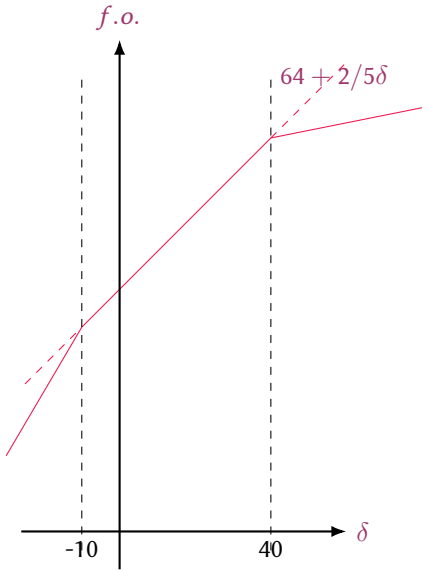
(II) Changes in RHS terms

| | $x_1$ | $x_2$ | $x_3$ | $x_4$ | $-z$ | $b$ |
|---|---|---|---|---|---|---|
| $x_3$ | 5 | 10 | 1 | 0 | 0 | $60+\delta$ |
| $x_4$ | 4 | 4 | 0 | 1 | 0 | $40+\epsilon$ |
| | 6 | 8 | 0 | 0 | 1 | 0 |

| | $x_1$ | $x_2$ | $x_3$ | $x_4$ | $-z$ | $b$ |
|---|---|---|---|---|---|---|
| $x_2$ | 0 | 1 | $1/5$ | $-1/4$ | 0 | $2+1/5\delta-1/4\epsilon$ |
| $x_1$ | 1 | 0 | $-1/5$ | $1/2$ | 0 | $8-1/5\delta+1/2\epsilon$ |
| | 0 | 0 | $-2/5$ | $-1$ | 1 | $-64-2/5\delta-\epsilon$ |

(It would be more convenient to augment the second. But let's take $\epsilon = 0$.)
If $60+\delta \implies$ all RHS terms change and we must check feasibility
Which are the multipliers for the first row? $k_1 = \frac{1}{5}, k_2 = -\frac{1}{4}, k_3 = 0$
I: $1/5(60+\delta) - 1/4 \cdot 40 + 0 \cdot 0 = 12 + \delta/5 - 10 = 2 + \delta/5$
II: $-1/5(60+\delta) + 1/2 \cdot 40 + 0 \cdot 0 = -60/5 + 20 - \delta/5 = 8 - 1/5\delta$
Risk that RHS becomes negative
Eg: if $\delta = -10 \implies$ tableau stays optimal but not feasible $\implies$ apply dual simplex

# Graphical Representation



(III) Add a variable

$$\begin{array}{rrcrcrcl} \max & 5x_0 & + & 6x_1 & + & 8x_2 & & \\ & 6x_0 & + & 5x_1 & + & 10x_2 & \leq & 60 \\ & 8x_0 & + & 4x_1 & + & 4x_2 & \leq & 40 \\ & & & & & x_0, x_1, x_2 & \geq & 0 \end{array}$$

Reduced cost of $x_0$? $c_j + \sum \pi_i a_{ij} = +1 \cdot 5 - \frac{2}{5} \cdot 6 + (-1)8 = -\frac{27}{5}$

To make worth entering in basis:

- increase its cost coefficient
- decrease the technological coefficient in constraint II: $5 - 2/5 \cdot 6 - a_{20} > 0$

(IV) Add a constraint

$$\begin{array}{lrcrcl} \max & 6x_1 & + & 8x_2 & & \\ & 5x_1 & + & 10x_2 & \leq & 60 \\ & 4x_1 & + & 4x_2 & \leq & 40 \\ & 5x_1 & + & 6x_2 & \leq & 50 \\ & & & x_1, x_2 & \geq & 0 \end{array}$$

Final tableau not in canonical form, need to iterate with dual simplex

| | $x_1$ | $x_2$ | $x_3$ | $x_4$ | $x_5$ | $-z$ | $b$ |
|---|---|---|---|---|---|---|---|
| $x_2$ | 0 | 1 | $1/5$ | $-1/4$ | | 0 | 2 |
| $x_1$ | 1 | 0 | $-1/5$ | $1/2$ | | 0 | 8 |
| | 0 | 0 | $-1/5$ | $-1$ | 1 | 0 | $-2$ |
| | 0 | 0 | $-2/5$ | $-1$ | 0 | 1 | $-64$ |

(V) change in a technological coefficient:

| | $x_1$ | $x_2$ | $x_3$ | $x_4$ | $-z$ | $b$ |
|---|---|---|---|---|---|---|
| $x_3$ | 5 | $10+\delta$ | 1 | 0 | 0 | 60 |
| $x_4$ | 4 | 4 | 0 | 1 | 0 | 40 |
| | 6 | 8 | 0 | 0 | 1 | 0 |

- first effect on its column
- then look at $c$
- finally look at $b$

| | $x_1$ | $x_2$ | $x_3$ | $x_4$ | $-z$ | $b$ |
|---|---|---|---|---|---|---|
| $x_2$ | 0 | $(10+\delta)1/5+4(-1/4)$ | $1/5$ | $-1/4$ | 0 | 2 |
| $x_1$ | 1 | $(10+\delta)(-1/5)+4(1/2)$ | $-1/5$ | $1/2$ | 0 | 8 |
| | 0 | $-2/5\delta$ | $-2/5$ | $-1$ | 1 | $-64$ |

# Relevance of Sensistivity Analysis

- The dominant application of LP is mixed integer linear programming.
- In this context it is extremely important being able to begin with a model instantiated in one form followed by a sequence of problem modifications
  - row and column additions and deletions,
  - variable fixings

  interspersed with resolves

# Summary

- Derivation:
  1. economic interpretation
  2. bounding
  3. multipliers
  4. recipe
  5. Lagrangian
- Theory:
  - Symmetry
  - Weak duality theorem
  - Strong duality theorem
  - Complementary slackness theorem
- Economic interpretation
- Dual Simplex
- Sensitivity Analysis